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Berlin School of Economics and Law

Master Digitale Transformation

Modul Digitale Technologien

Introduction Digitization, Trends, History

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Security and Embedded Systems Engineering

What is Digitization?

- Literally idea: Transformation of analog signals into digital ones, i.e.,
  - Finding digital representations for analog information, objects, or events
  - Storing digital information, distribute them, and process them
- nowadays synonymous for digital revolution, transformation
- Digital change: change processes in society including economy, culture, education, and politics
- Advantages of Digitization
  - Enables digital, automated processing
  - Machine-based readability and processability

Companies live in an ever-changing and evolving environment where digitization is key

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- 52% of Fortune 500 have vanished since 2000
- Mean life expectancy of a company: in 1958: 61 years, in 2015: 15 years
- Examples for disrupted companies:
  - Blockbuster (1985-2010): Home video rental service: 2004 84.300 employees, 9.094 branches, 2000 Netflix offered to sell ist business to Blockbuster for 50 Mio. $, CEO was not interested because he considered it to be a “very limited niche business” → underestimated effects of digitization
  - Polaroid (1937-2001): Instant cameras, 2001 insolvence → underestimated effects of digitization (digital photography)
  - Kodak (1981-2012): various experiments with digital photography, however, never fully supported due to fear of undermining the traditional business; ofoto 2001 bought by Kodak, but instead of developing something like Instagram, used to print pictures
  - Toys R Us (1948-2017): did not develop e-commerce, 2017 (too late) 100 Mio.$ investment to achieve a turn around and develop an E-Commerce shop → underestimated effects of digitization
  - Tower Records (1960 – 2004): Music store (CD, cassettes, DVD etc.) → no match against Spotify, iTunes, and piracy

Quelle: https://www.linkedin.com/pulse/since-2000-52-companies-fortune-500-have-vanished-sam-tanham/, 2021

Spectacular Misjudgments



Social Change in the Digital Era



Some Challenges for Digitization

Heterogeneous Architectures
- Many different technologies
- Badly documented code and interfaces
- Architectural issues and insufficient implementation of non-functional requirements (e.g. modularization, extandability)

Fast changes in technology
- Computer Science underwent a tremendous change in recent past:
  - Software Engineering has changed tremendously
  - New technologies (e.g. Cloud Computing, Artificial Intelligence, Data Science / Big Data) have risen or changed increasingly
  - New technologies and demands lead to more complex requirements → skills and experience required
- Lack of standardization and quich changes paralyze

Domain Knowledge

Apart from technology, deep domain knowledge is necessary

Communication
- For n communication participants there are $\frac{n \cdot (n+1)}{2}$ channels
- ‘grown‘ structures complicate changes
- Intransparent knowledge structures in companies

Difficulties related to changes
- Heterogeonous or lack of skills
- Change aversion

Some ”Laws”

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1. „Moore‘s Law“
2. “Neven‘s Law“
3. „Amdahl‘s Law“
3. „Gilder‘s Law“
4. „Metcalfe‘s Law“



Bildquelle:
https://www.funds-europe.com/june-2020/sponsored-feature-pandemic-boosts-digitalisation

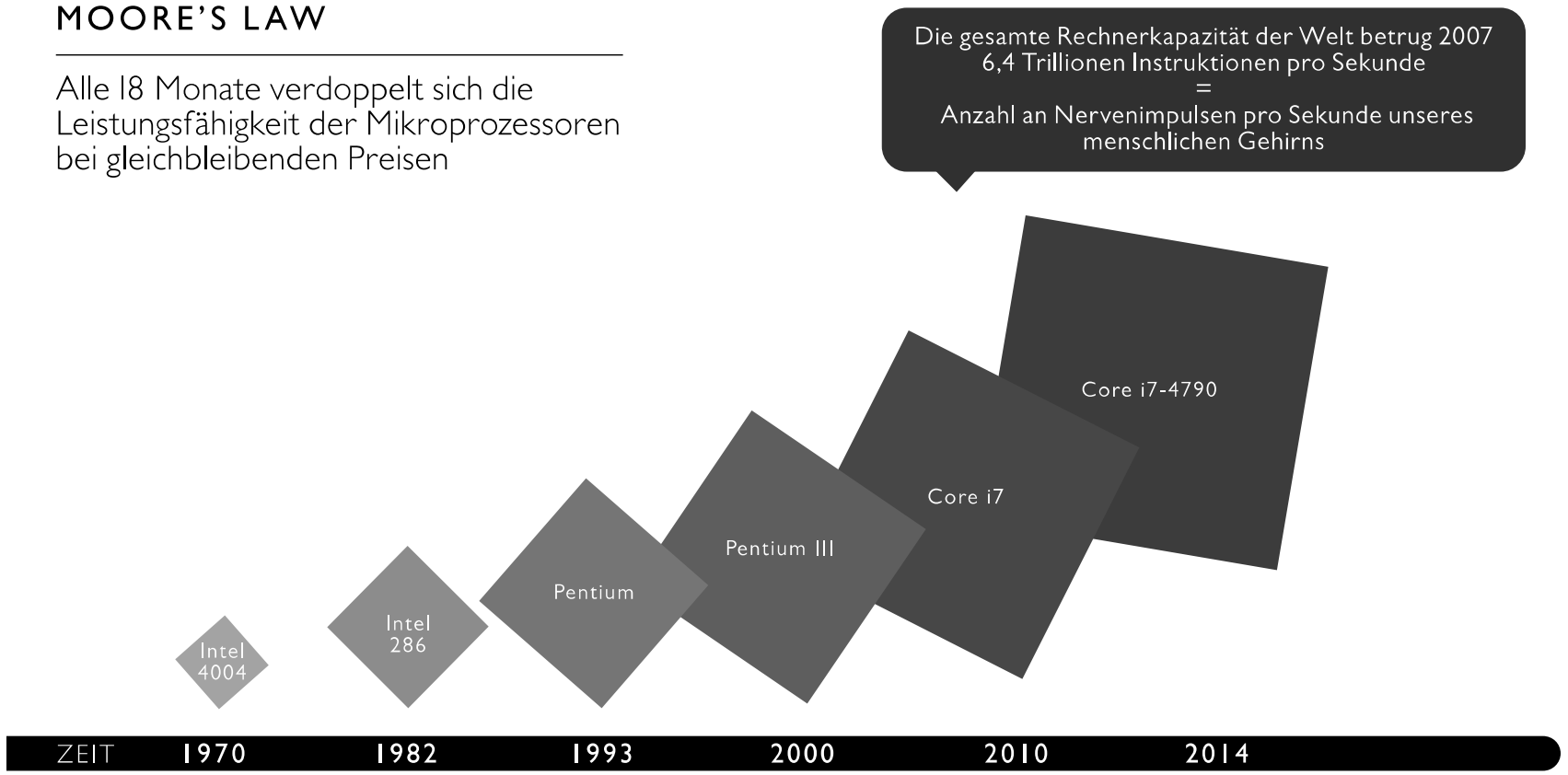
Moore's Law

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Moore's Law is the observation that the number of transistors on an integrated circuit will double about every two years with minimal rise in cost

Effectively:

- More transistors → more functionality, higher performance (in many cases)
- Lower cost per transistor, over time
- Smaller geometry, finer lithography, improved yield and manufacturing techniques



Moore’s Law Discussion

1

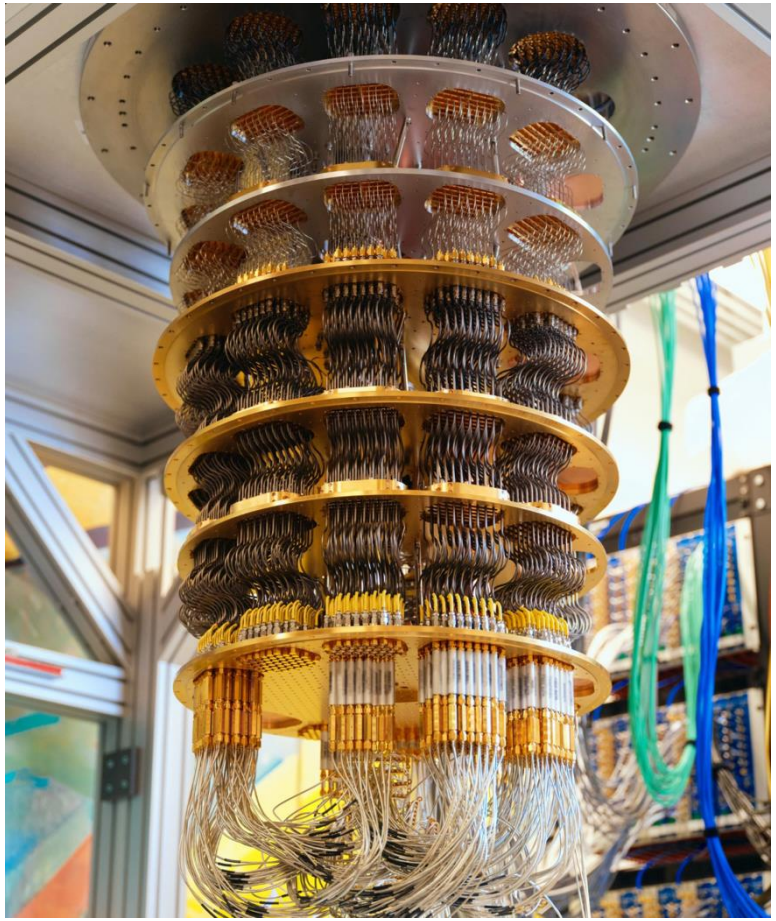
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- By 2025, Moore’s Law remains influential, though its relevance is challenged by the physical limitations of transistor miniaturization
- Issues
- Quantum effects: tunnelling effect, leakage current, short-channel effects become significant
- Heat dissipation / power density: more power per area, more heat to remove
- Lithography / manufacturing complexity: The equipment to pattern ever-smaller features becomes extraordinarily expensive (EUV lithography, extreme ultraviolet; mask costs; yield issues)
- Diminishing returns: Even if transistor count doubles, actual single-thread performance, or cost per performance, may no longer improve as much. Many systems shift to parallelism.
- Dennard Scaling stopped working: says that as transistors get smaller, they get more energy-efficient; 2005-2010 it turned out: reducing voltage causes failures, voltage kept constant while transistor count kept rising

Neven's Law

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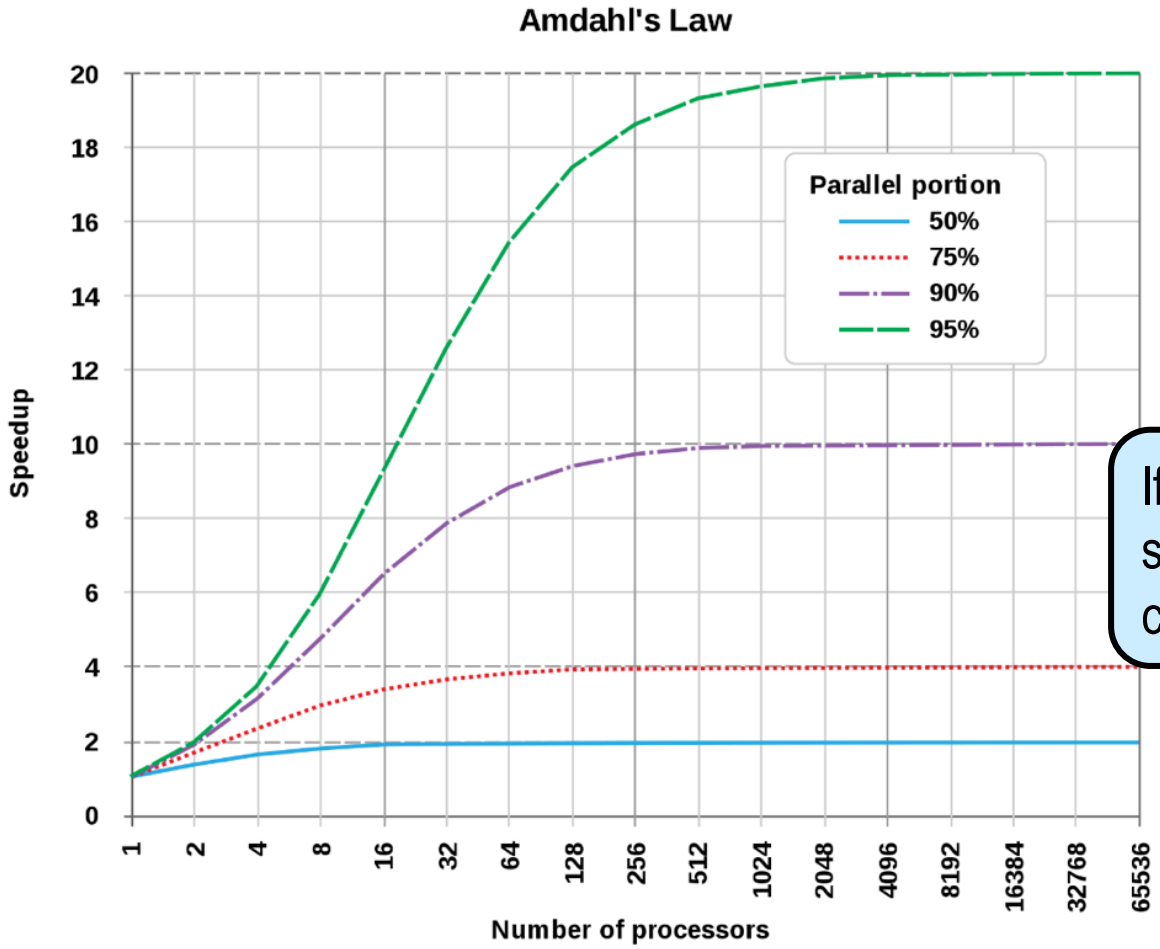
- The performance of quantum computers is improving at a doubly exponential rate relative to classical computers. (Hartmut Neven, director of Google's Quantum AI Lab, 2019)
- Discussion
- Moore's Law: performance 2^n, Neven's Law: 2^{2^n}
- HOWEVER
  - Only observation in early stages; heuristics rather than physical law
  - Observation: stable qubits increase exponentially and algorithmic improvements likely
  - In 2025, this does not hold completely:
    - Progression in stable qubits not as exponential as projected
    - Algorithms – mainly improvements but not many breakthroughs like Shor's Algorithm
- Quantum Computing: will have huge impact on specific problem solutions (e.g. prime factoring, simulations of physical environments) but not on classical computing in general



Source: https://quantumai.google/discover/whatisqc

Parallel execution: Amdahl's Law

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If 50% is sequential, the maximum speedup is 2, no matter how many cores you use



Gilder's Law

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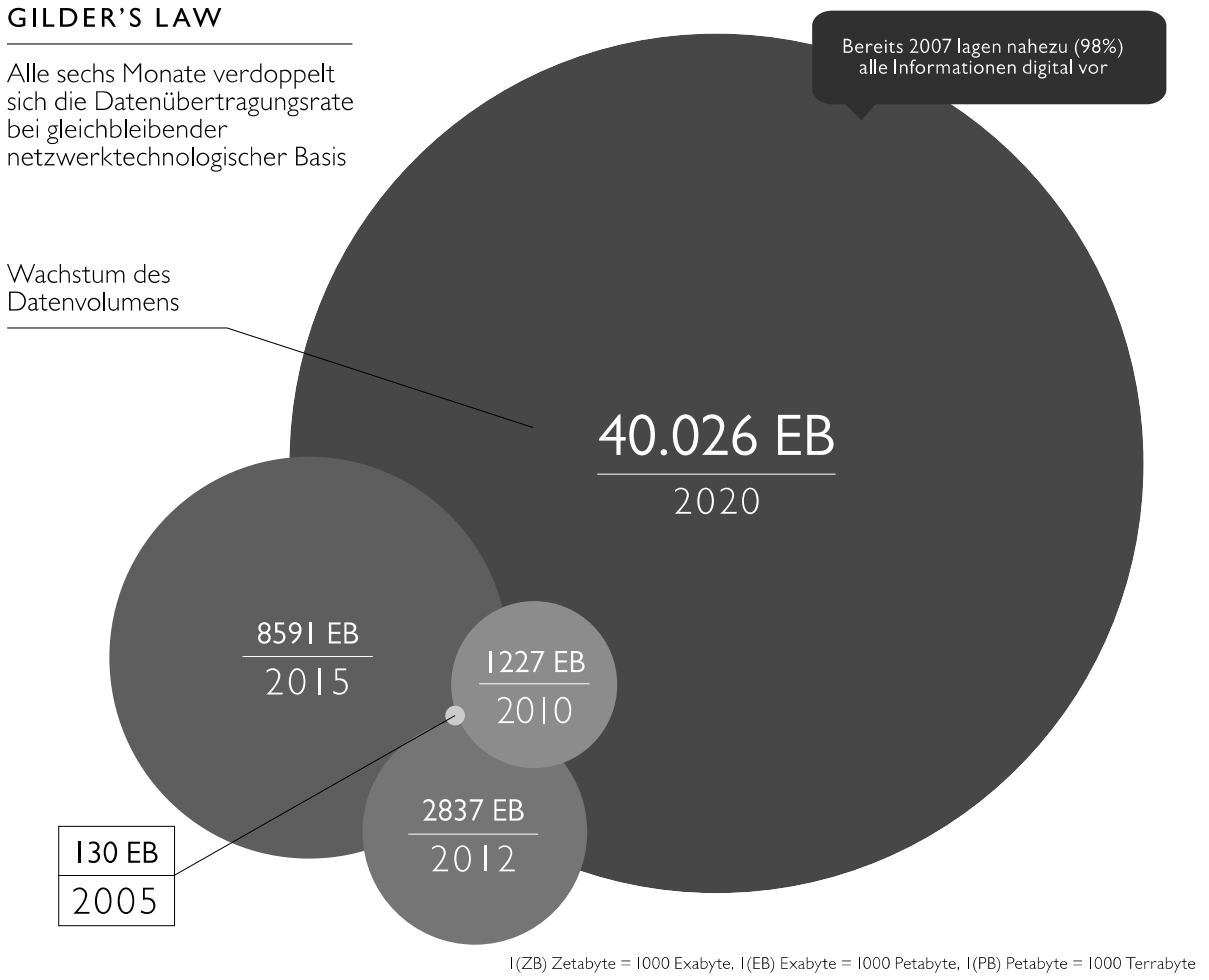
The total bandwidth of communication systems triples every twelve months

or

The bandwidth of communication networks grows at least three times faster than computing power

So:

- Moore's Law → CPUs and chips get faster (computing side).
- Gilder's Law → Networks get broader and faster (communication side)

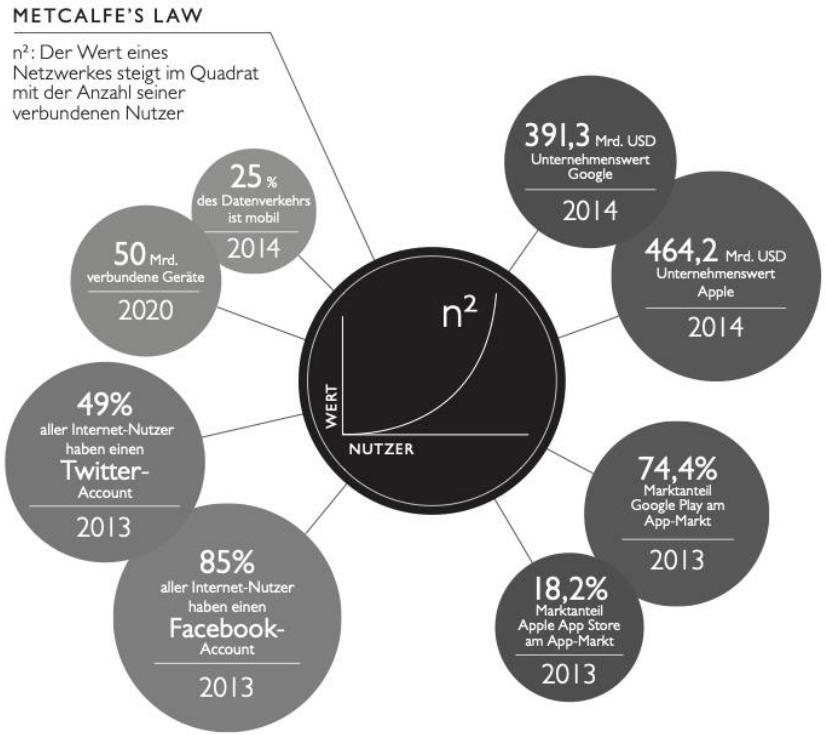


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Metcalfe's Law

- The value of a network is proportional to the square of the number of its connected users.

- Central idea: Connections between users grows quadratically
- Helps explain:
- Network effects — why networks become exponentially more valuable as they grow
- The dominance of platforms like Facebook, LinkedIn, X (Twitter), and WhatsApp: once a network reaches critical mass, its value to each user increases so much that it's hard for competitors to catch up
- The rapid growth of the Internet and social networks in the 1990s–2000s.

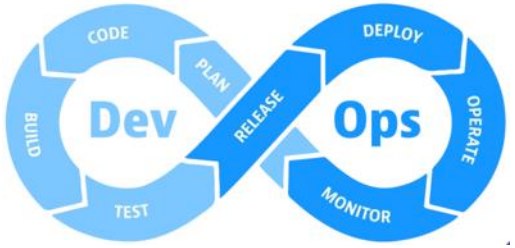


Overview of some developments

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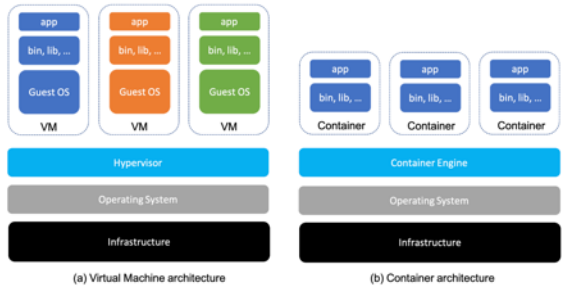
Programming languages and developments

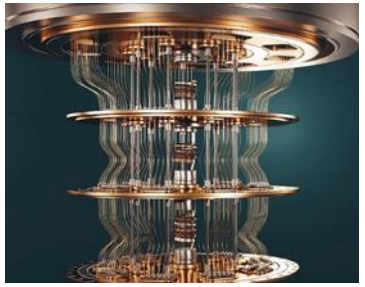
Software Engineering Developments





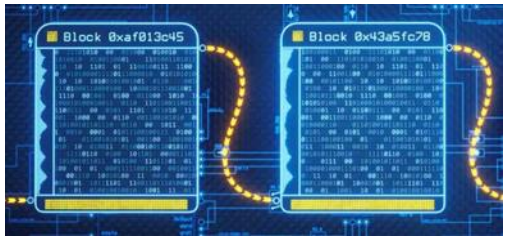
virtualization



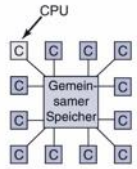


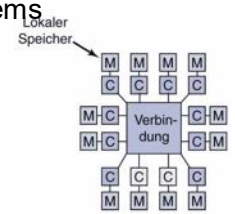
Quantum Computing

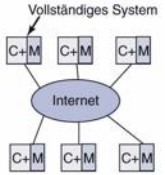
Block Chain



Multiprocessor and distributed systems



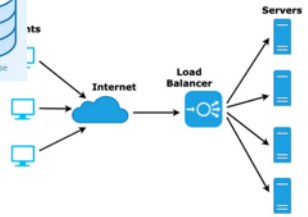


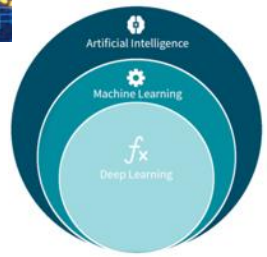










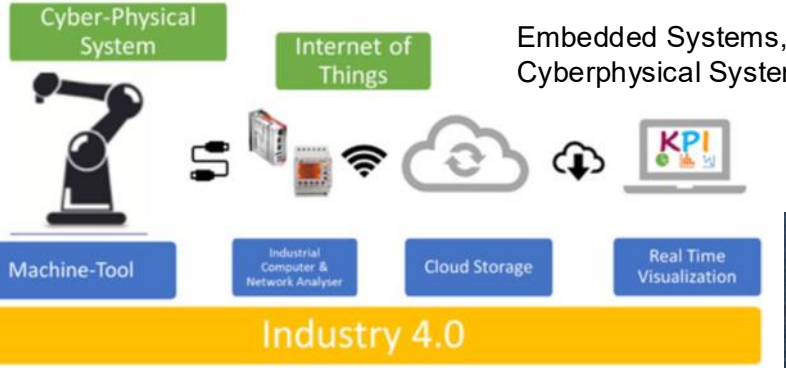






Apache Flink

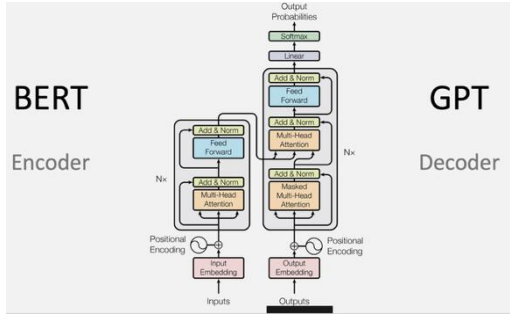
distributed processing, distributed storage, stream processing



Embedded Systems, Cyberphysical Systems



Machine Learning

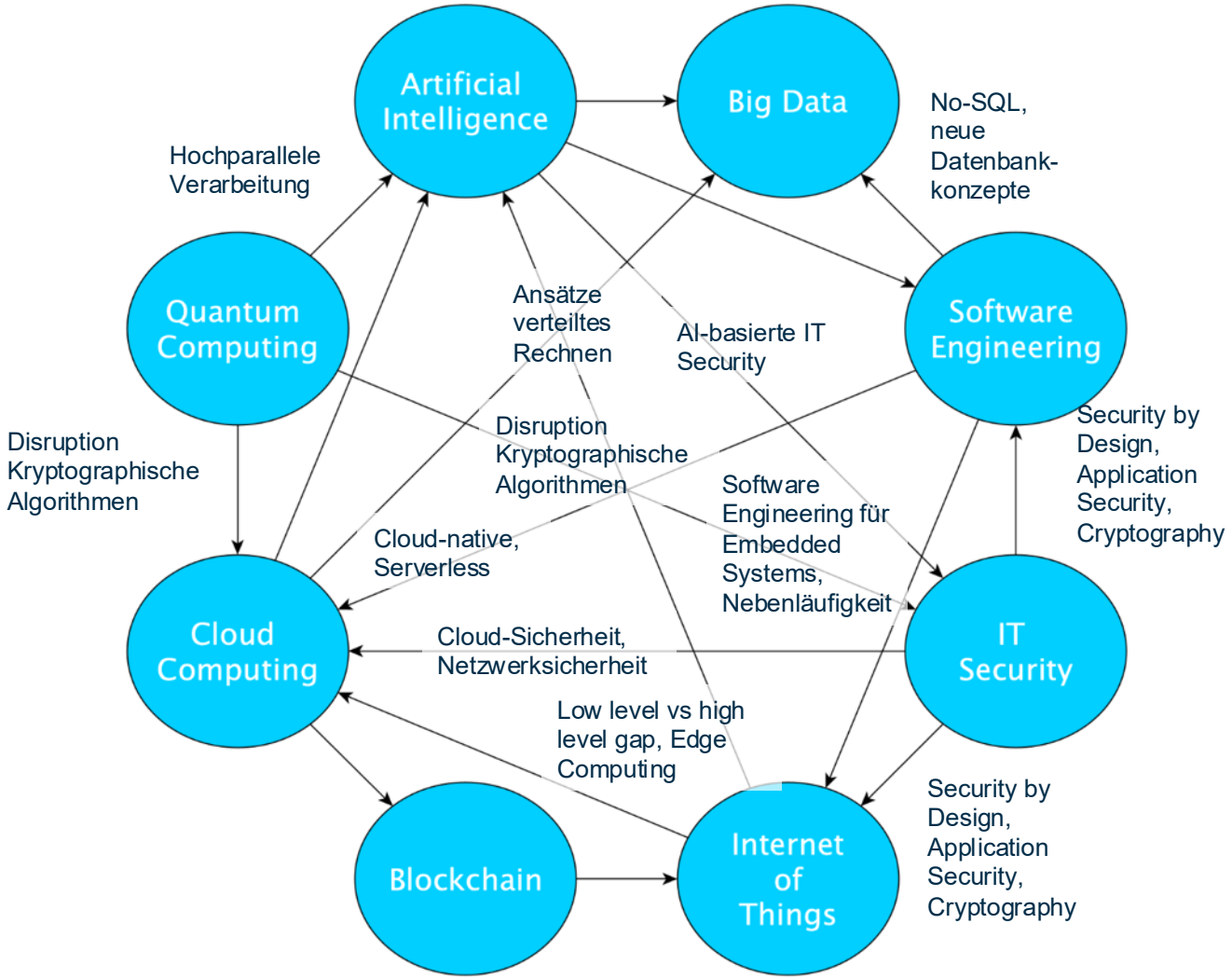


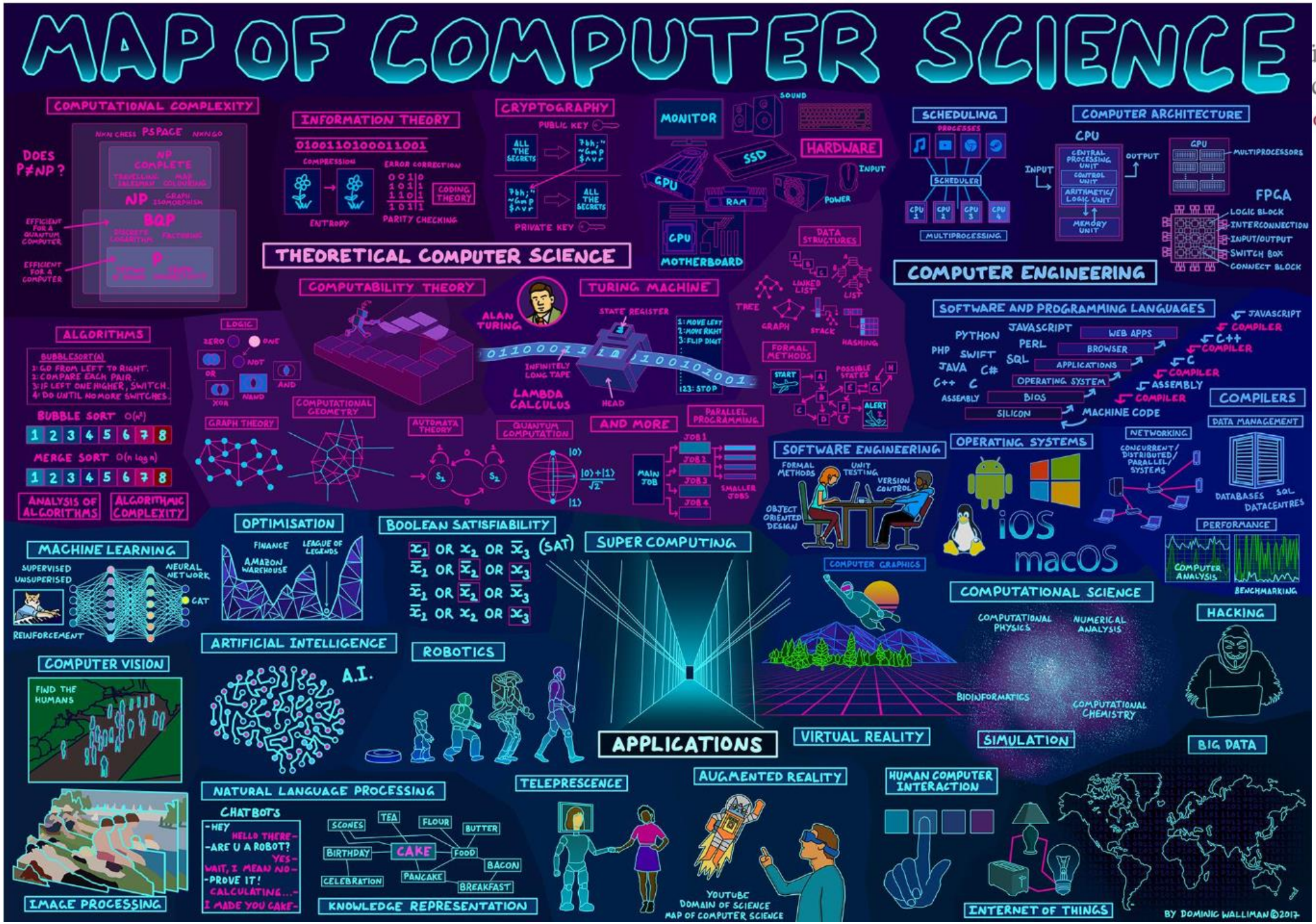
Cloud Computing



Interdependencies between developments

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Quelle: https://monadical.com/posts/map-of-cs-subfields.html

Brief Group Discussion

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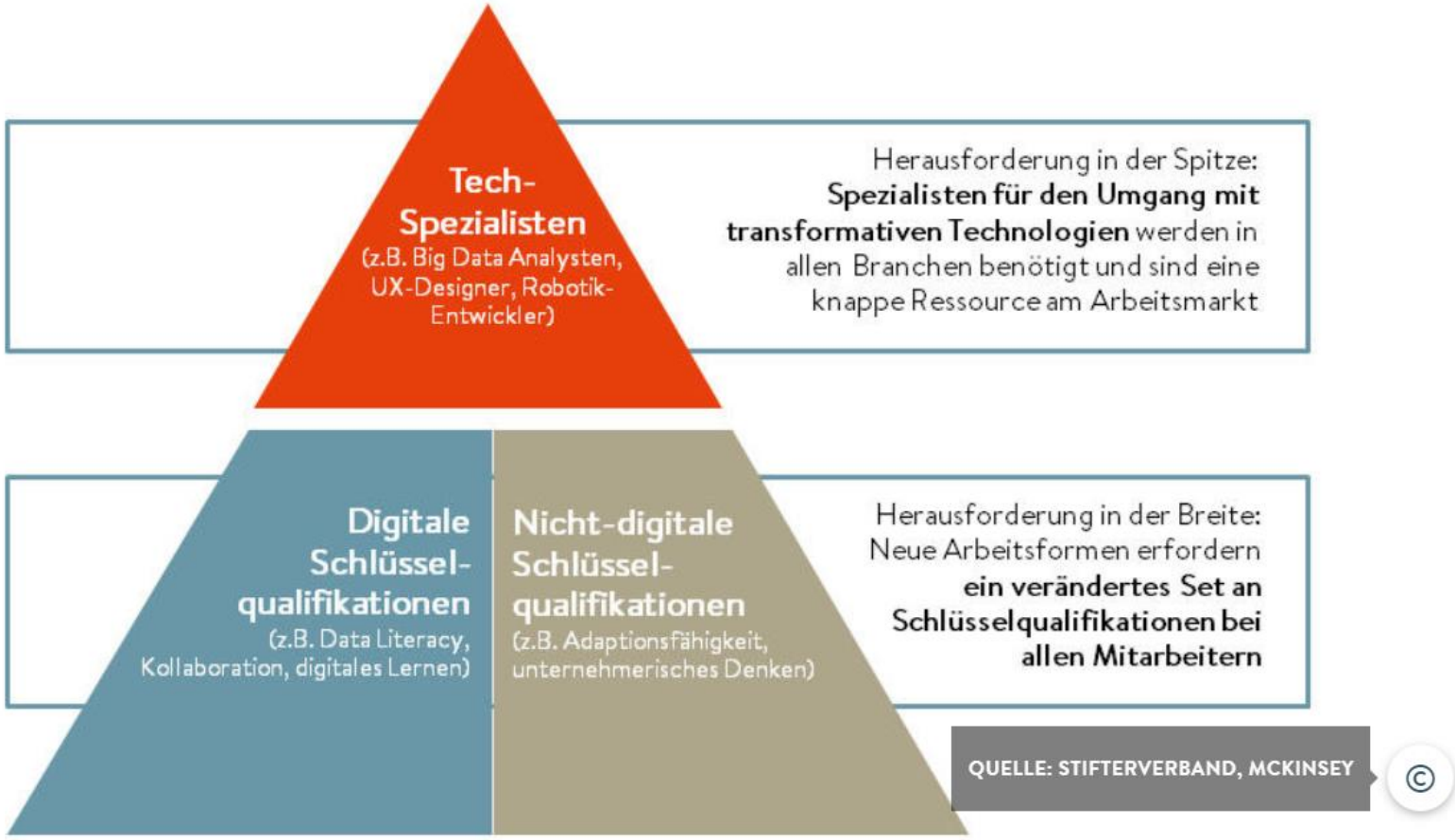
- Reflect some personal experiences with the laws, the tech developments we just went through and discuss their impact on economy and society.

- 20 mins discussion in groups, 1 group presents, feedback from the audience

Future Skills

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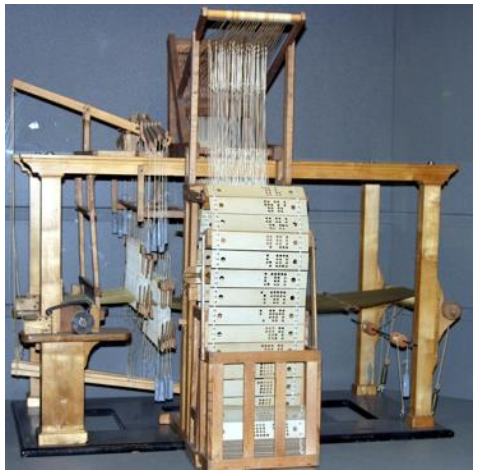
- What do you think? Which skills will be most relevant in the future?
- Also reflect on that (10 mins) and we discuss it.



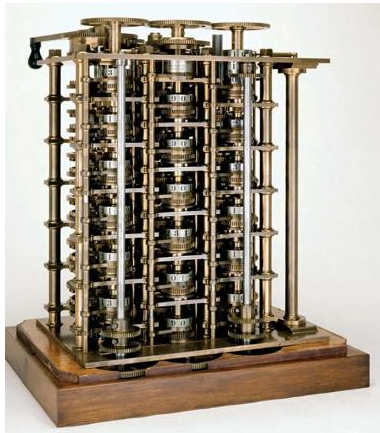
Short history of the computer

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- 1785 Edmund Cartwright first mechanical loom, enhancement 1805 by Jacquard (punchcards)



- 1812 Charles Babbage invents Difference Engine (1833 built by Joseph Clement)
- 1830 Charles Babbage Plans for Analytical Engine (arithmetic operations, punchcards, storage, sequential control)



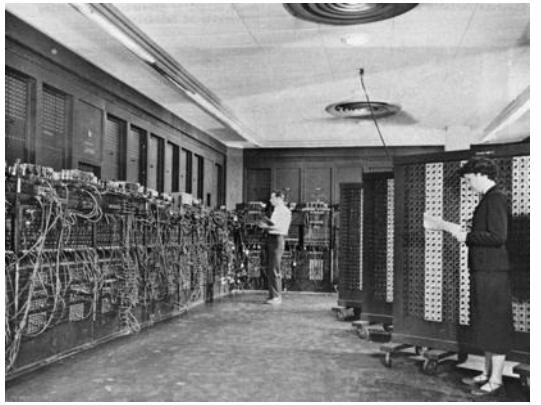
- 1843 Ada Lovelace writes first computer program for Analytical Engine
- 1938 – Z1 Konrad Zuse (binary numbers, coding on pierced cinema
cards)



Short history of the computer

- 1941 Konrad Zuse builds Z3, 1943 IBM boss: “I believe there is a worldwide demand for perhaps 5 computers.“, 1945 Z4
- 1946 Eckert and Mauchly build the “Electronic Numerical Integrator and Computer” (ENIAC) for US governments, first universal computer
- 1965 Moores‘ law
- 1970 Texas Instruments develops microprocessor
- 1977 Ken Olson (DEC): „There is no reason for somebody having a computer at home.“







Short history of the computer

- 1980er flowering period of home computers: C64, Amiga
- 1981: IBM-PC
- 1981: MS DOS
- 1982: 80286 processors, 1986: 80386, 1989: 80486
- 1990er: internet and WWW
- 2005: first Dual-Core processors, 2007 four cores









Short history of the computer

- 2000er dotcom bubble
- 2002 AWS kicks off

- 2008 Google Cloud Platform kicks off
- 2010 Microsoft Azure kicks off

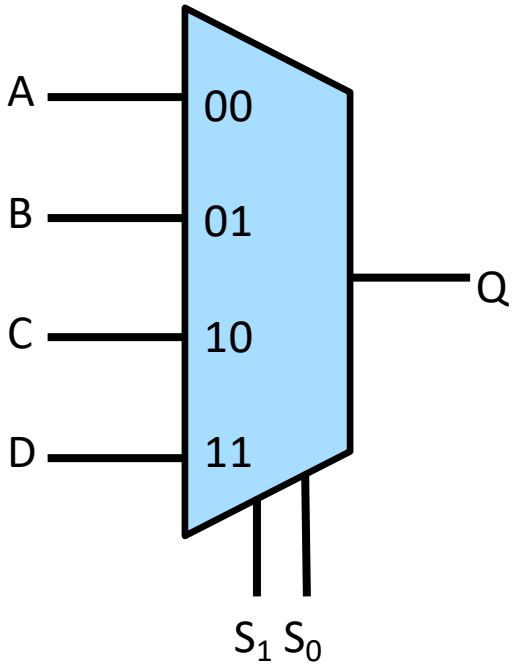
- 2016 IIoT Vision, Data Science, AR / VR, Security are developed or receive increased attention
- 2016 Alpha Go beats best Go player of the world

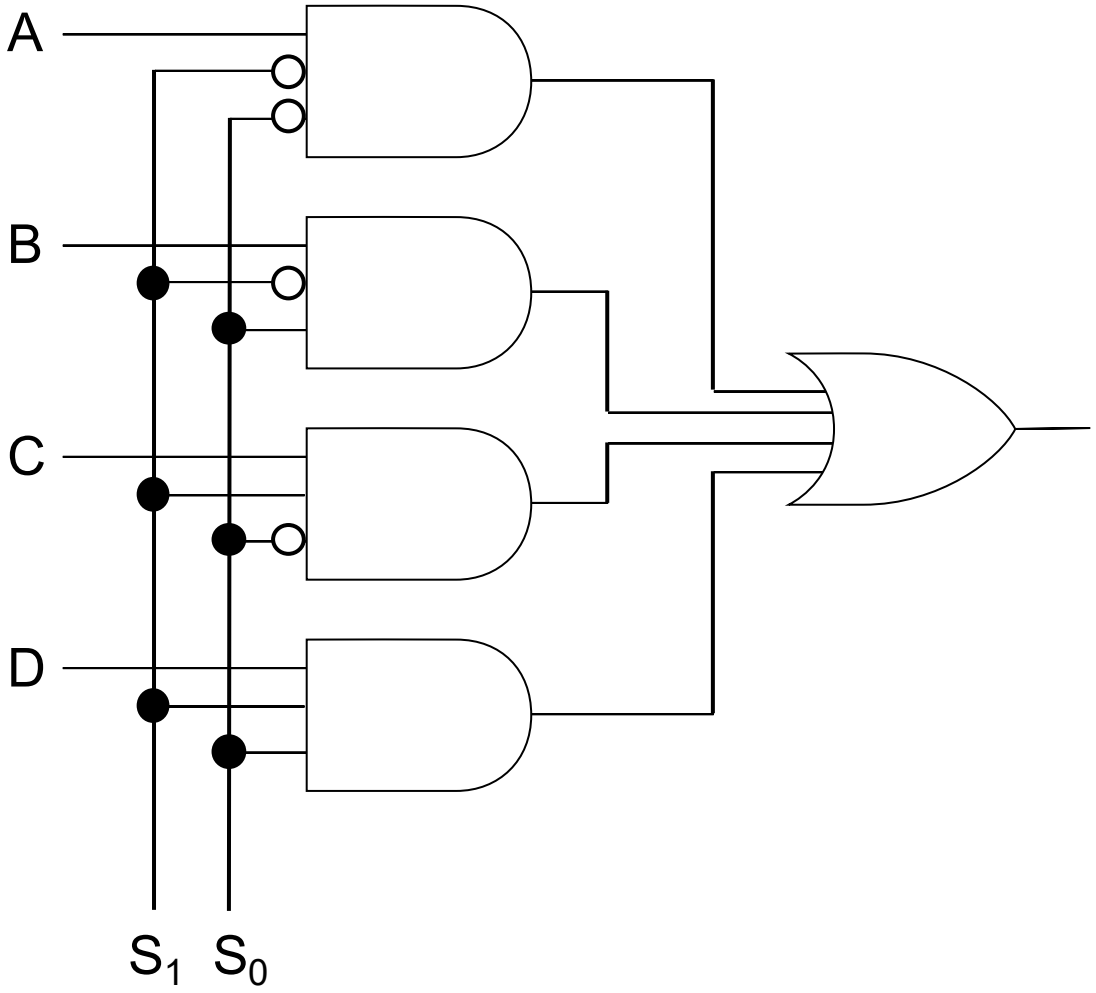
- 2021 autonomous driving allowed on level 3



Foundational concepts of computers: digital circuits

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Foundational concepts of computers: abstraction, assembler, compiler

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Hochsprache (z.B. C)

```
void swap(int v[], int k)
{
    int temp;
    temp = v[k];
    v[k] = v[k+1];
    v[k+1] = temp;
}
```

C compiler

Assembler

```
swap:
  muli $2,$5,4
  add  $2,$4,$2
  lw   $15,0($2)
  lw   $16,4($2)
  sw   $16,0($2)
  sw   $15,4($2)
  jr   $31
```

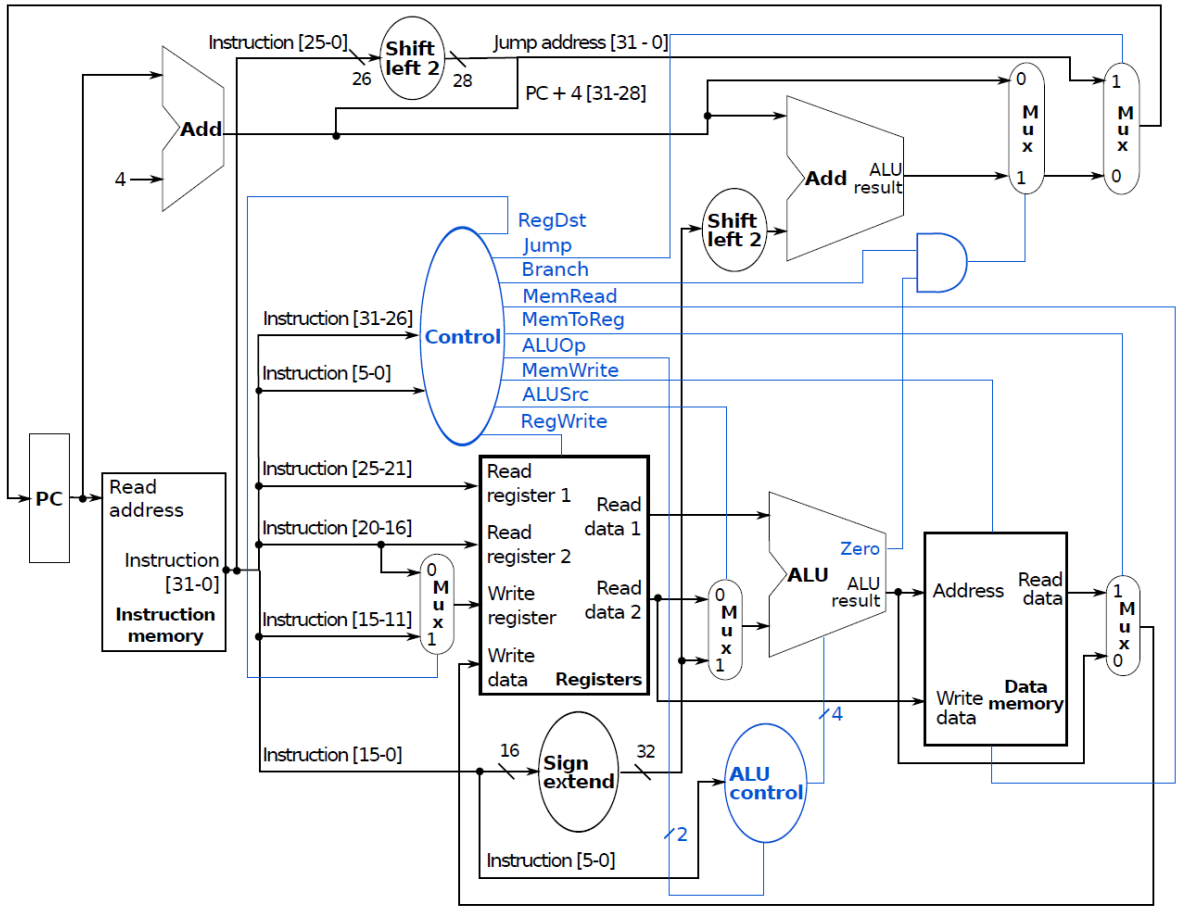
assembler

```
00000000101000010000000000011000
00000000100011100001100000100001
10001100011000100000000000000000
10001100111100100000000000000100
10101100111100100000000000000000
10101100011000100000000000000100
00000011111000000000000000001000
```

Maschinelle Binärsprache

Processors of Classical Computers

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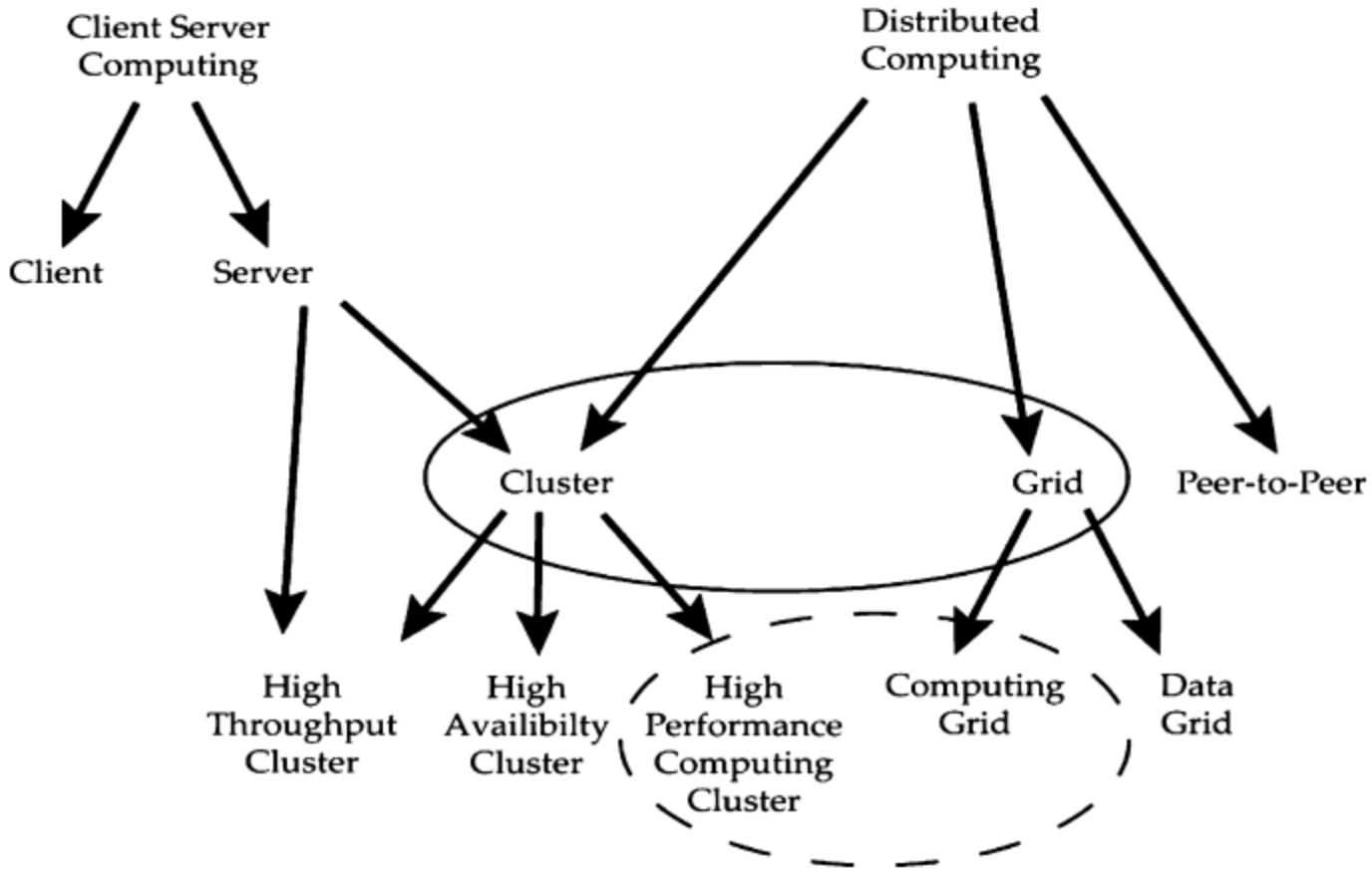




John von Neumann

Von Neumann Architecture:
Computer as Meta-Machine: Data and program both in main storage

Distributed Systems



Quelle: Bengel et.al. Masterkurs Parallele und Verteilte Systeme, 2. Aufl., Springer 2015

From Distributed Computing to Cloud Computing

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IaaS
(infrastructure as a service)

PaaS
(platform as a service)

SaaS
(software as a service)

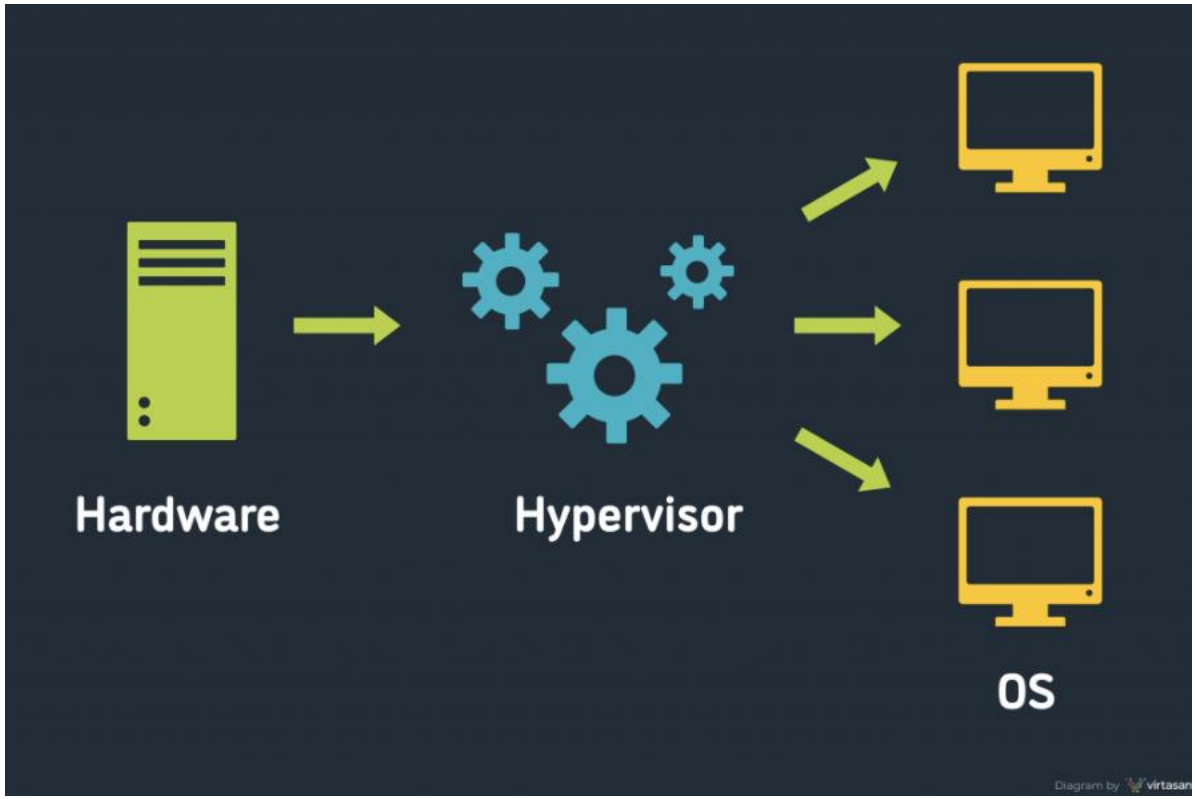


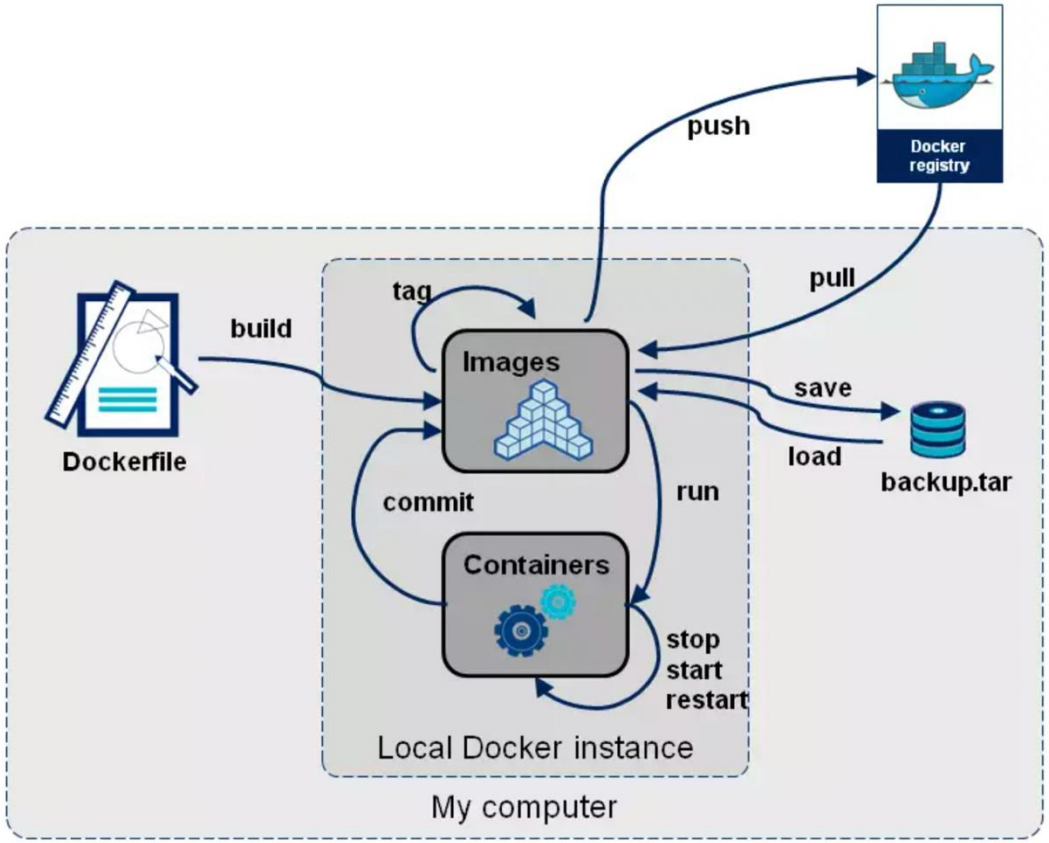
Cloud computing is the on-demand provision of computing power, databases, storage, applications and other IT resources via the Internet at usage-based prices.

2002 – AWS kicks off

aws

Virtualization and Containerization

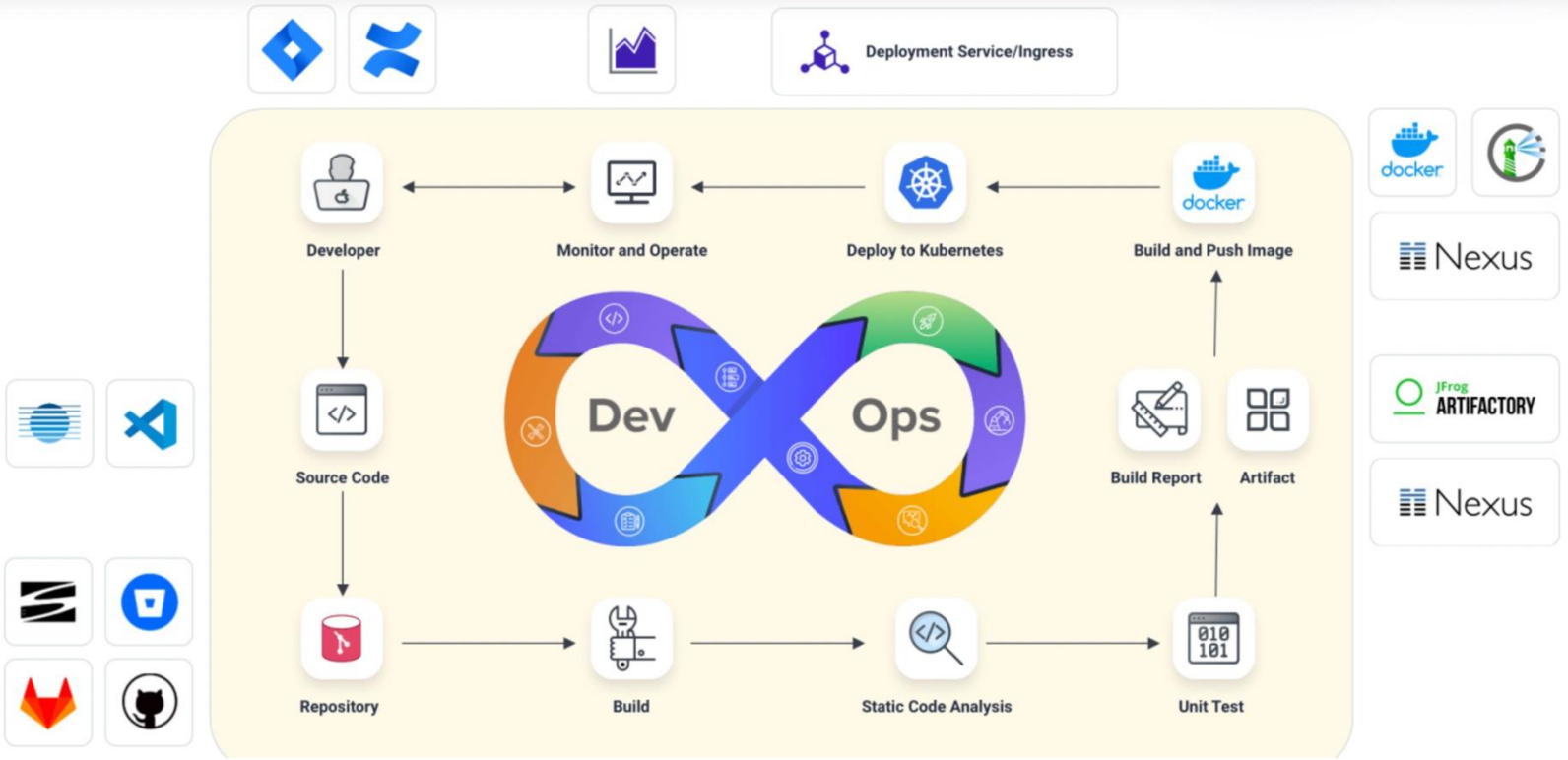




DevOps

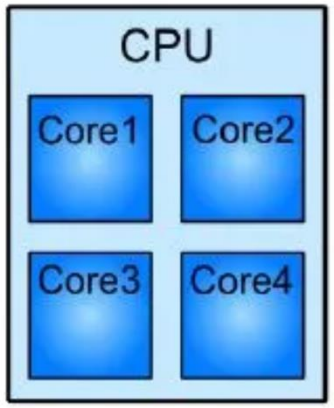
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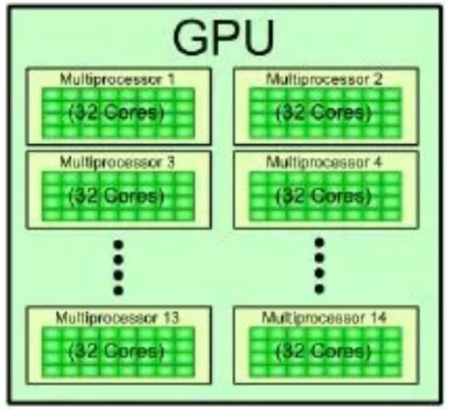
- DevOps is a set of practices, cultural philosophies, and tools that integrate software development (Dev) and IT operations (Ops) to improve collaboration and enable faster, more reliable, and continuous delivery of software.

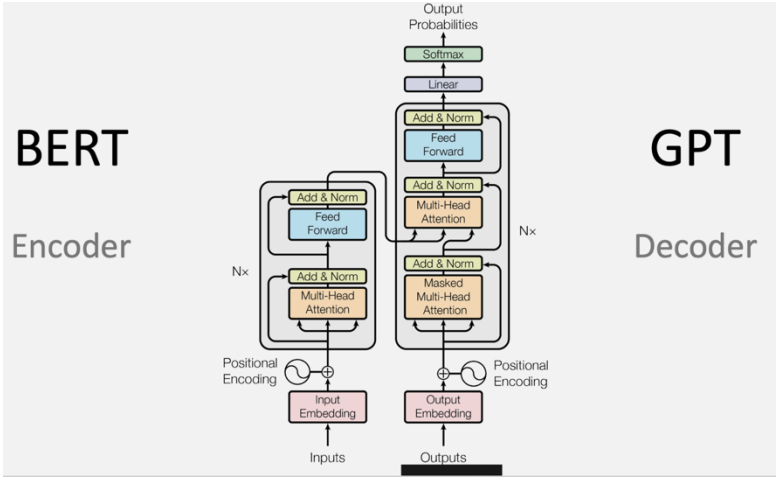


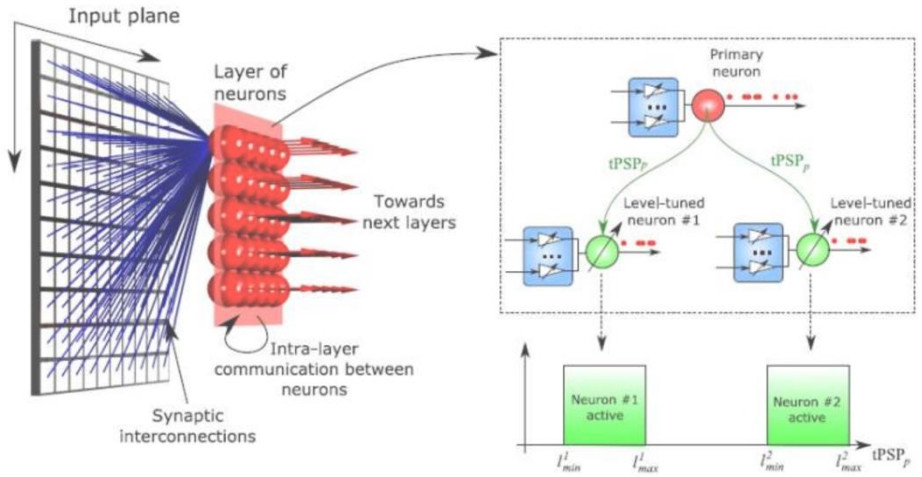
Machine Learning triggers new hardware developments

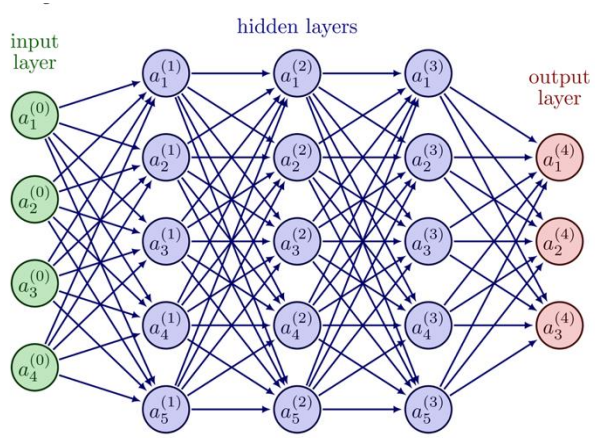
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Quelle: https://phys.org/news/2016-08-neuromorphic-mimics-important-brain-feature.html

Developments in storage

- 1890-1970 punchcards (comparison: mean SSD storage nowadays: 400GB, corresponds to ca. 5 billion punchcards)
- 1932 drum storage (hard drive predecessor), rotating ferromagnetic metal cylinderand multiple reading and writing heads
- 50s: magnet storage (1 coil : a couple of MB) → still used for backup
- 1956 first hard drive IBM 350: 1t mass, 5 MB storage
- 1965 100 MB hard drive
- 1969 disks start (8 inches)
- 1976 5,25 inch disks
- 1982 3,5 inch disks



Developments in storage

- 1989 first CD
- 1996 flash drive
- 2001 DVD & SD cards
- 2000er SSDs start displacing hard drives (80ern developed, first 3,5 inch SSD with 14 GB: 42.000 USD in 2001)
- 2006 BluRay
- 2011 NVMe (Non Volatile Memory Express) - more speed via PCI-Express instead of SATA for SSDs

Key Takeaways

Digitization enables automation and scale
Digital representations make information computable, reusable and shareable.

Computing evolves through capacity and abstraction
Hardware advances and software abstractions underpin today's connected systems.

Technology "laws" have limits
Growth trends need context. Sequential work limits the gains from parallel computing.

Technologies depend on each other
Cloud, AI and connected devices rely on suitable architectures and security.

Management decisions start with the use case
Technology choices must fit business value, cost, security and available skills.